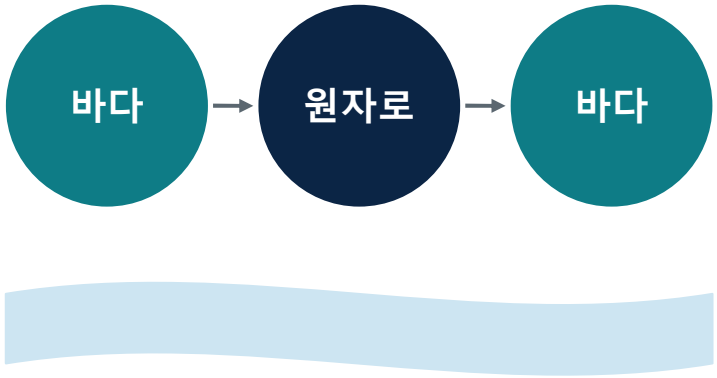
2026.11.13, 해양로봇기술연구회 초청강연

바다에서 원자로까지, 그리고 다시 바다로



원자력로봇과 해양로봇 기술의 접점
- 함께 풀 수 있는 문제는 어떤 것들이 있나?

주성문
한국원자력연구원 (KAERI)

한국원자력연구원

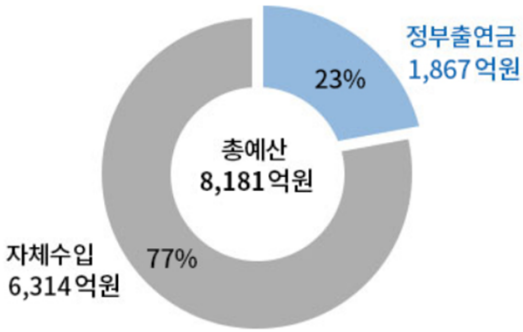
- 1959년 2월, 원자력연구소(국립 연구소)
- 1973년 2월, 한국원자력연구소(정부출연 연구소)
  - 원자력연구소, 방사선의학연구소,방사선농학연구소 통합
- 2007년 3월,한국원자력연구원으로 명칭변경
- 국내 유일의 원자력 종합 연구개발 기관



직원 현황 2026년 3월 1일 기준



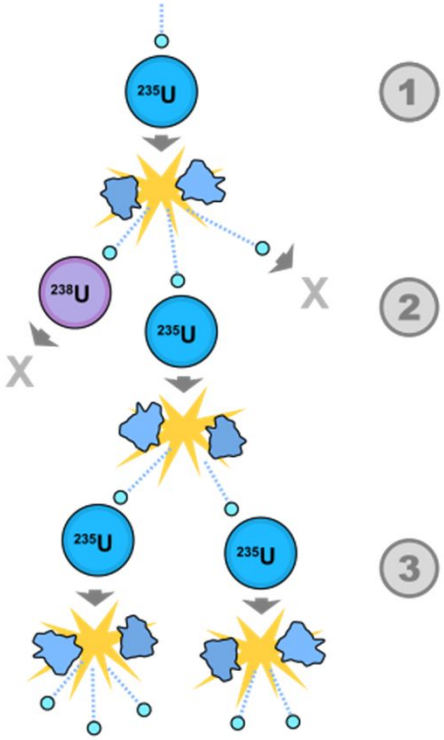
예산 2026년도 사업계획 및 예산(안) 기준



지역별 거점

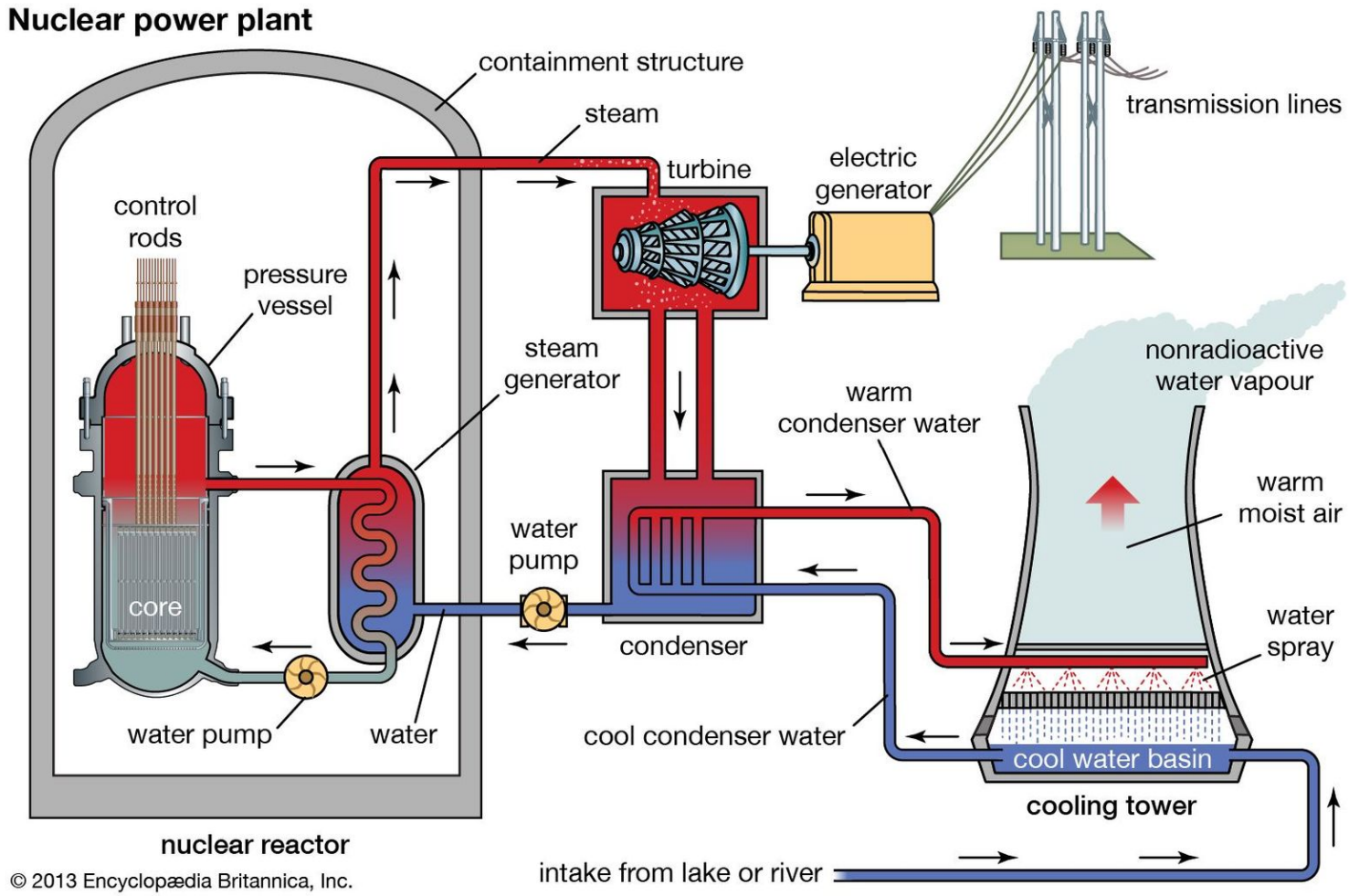


원자력

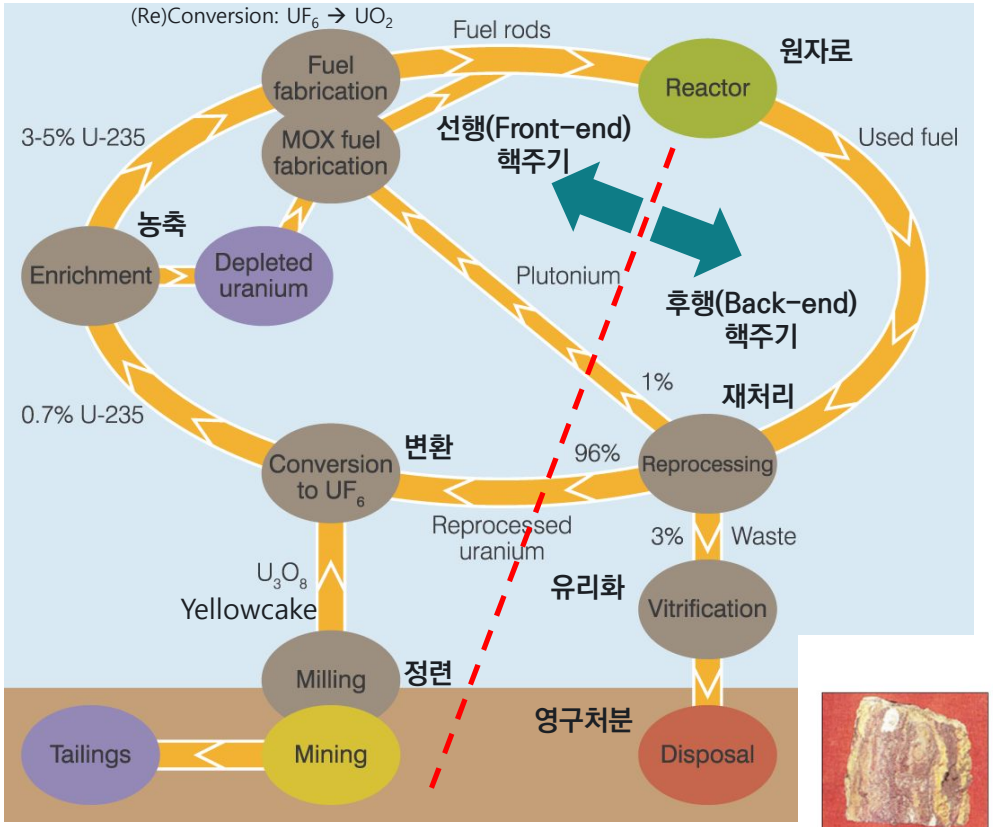

235U
1
238U
X
235U
X
2
235U
235U
3

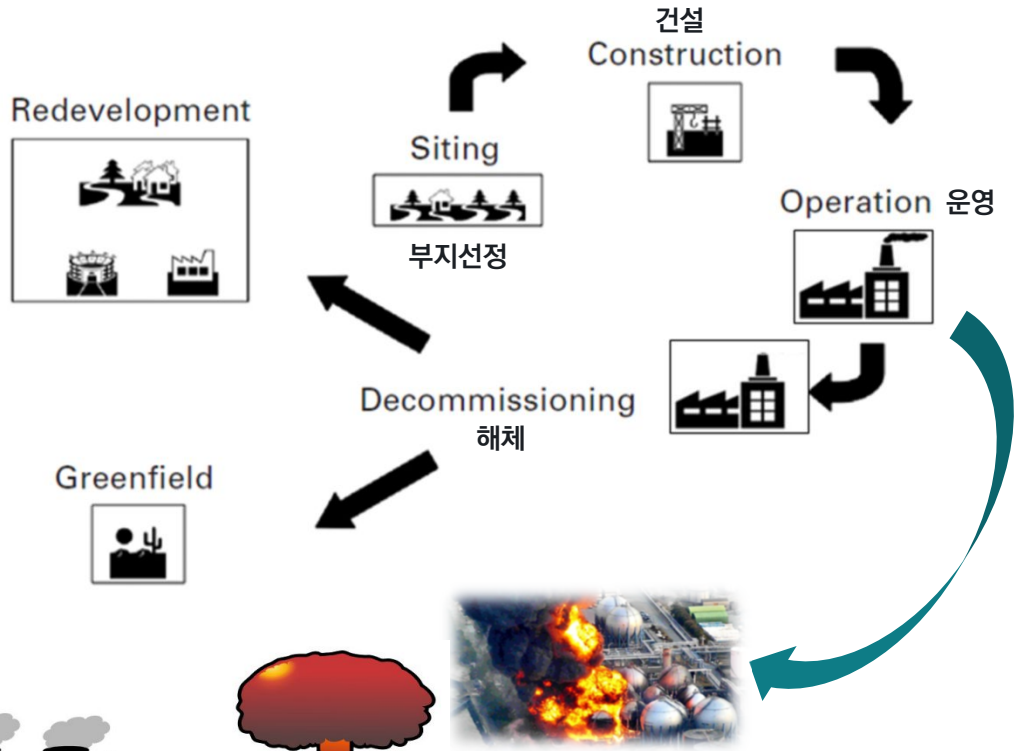

120 gallons
Oil
17,000 ft³
Natural Gas
Natural gas is less energy dense compared to other fossil fuels, but creates far fewer emissions.
A uranium pellet packs around 17 million BTU (British Thermal Units) of energy.
1 inch tall
Uranium Pellet
Less than 10 pellets can power the average household for a year.
1 ton
Coal
Source: U.S. Department of Energy
1
Uranium pellet
2
Cladding
3
Rod
4
Fuel assembly
10 mm
8 mm
≈ 4 m
≈ 4 m
≈ 25 cm

원자력 발전

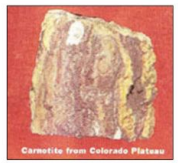

Nuclear power plant
containment structure
steam
turbine
electric generator
transmission lines
control rods
pressure vessel
steam generator
warm condenser water
nonradioactive water vapour
warm moist air
water pump
core
condenser
water spray
water pump
water
cool condenser water
cool water basin
cooling tower
nuclear reactor
intake from lake or river
© 2013 Encyclopædia Britannica, Inc.

원자력 주기(Cycle): 연료(Fuel), 시설(Facility)

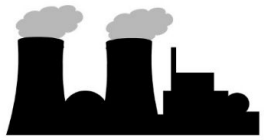




*event={deviation, incident, accident}



Natural Uranium:
0.7% ^{235}U
99.3% ^{238}U



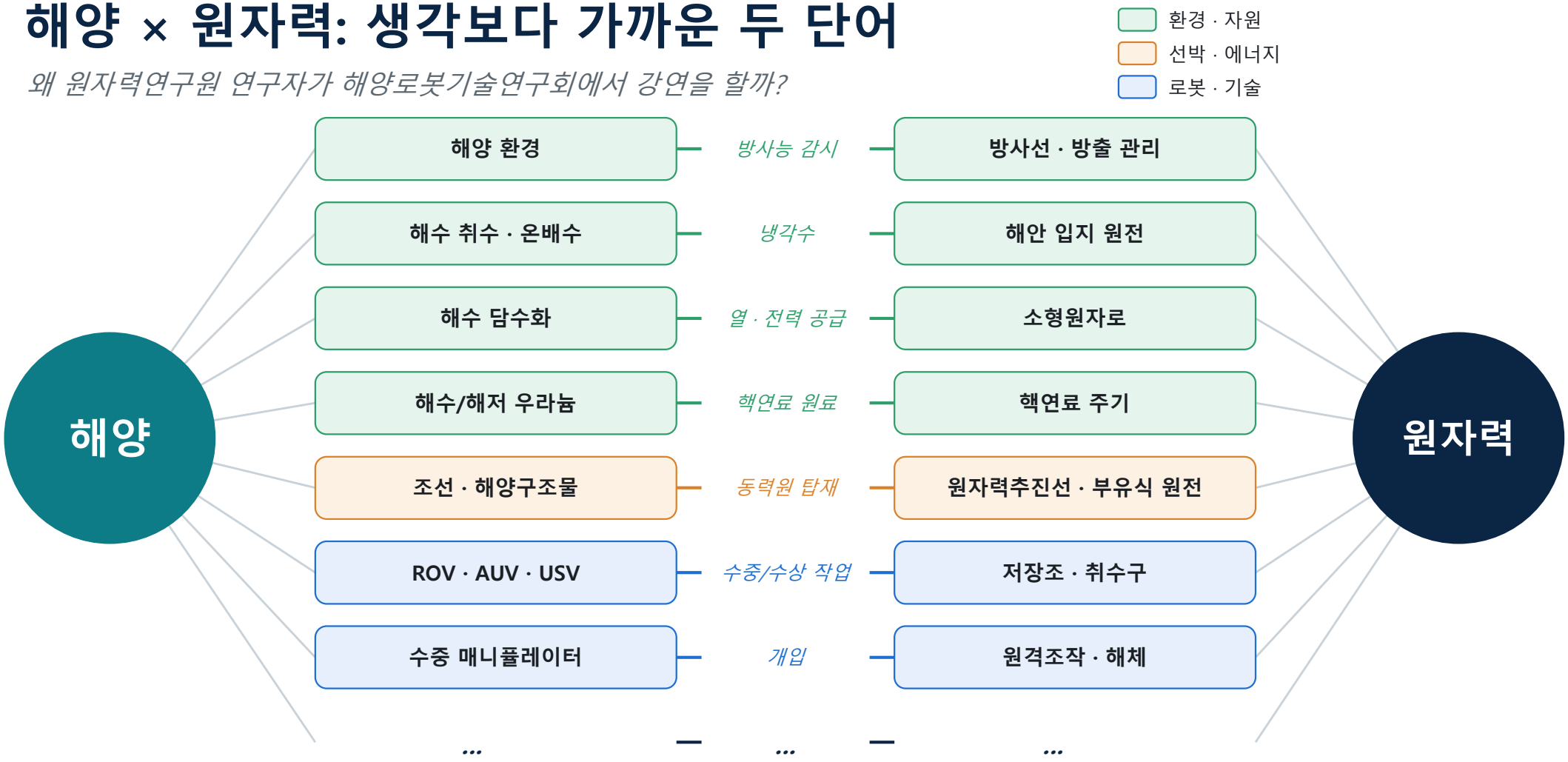
Low Enriched Uranium (LEU): 3%-5% ^{235}U
Fuel for Power Plants



Weapons-grade Uranium: >90% ^{235}U

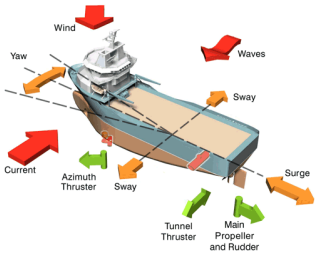
해양 × 원자력: 생각보다 가까운 두 단어

왜 원자력연구원 연구자가 해양로봇기술연구회에서 강연을 할까?

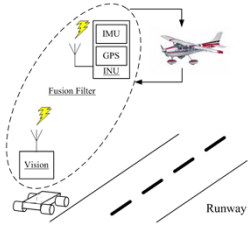


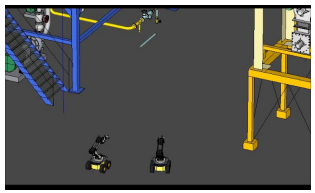
바다에서 → 원자로까지 → 다시 바다로

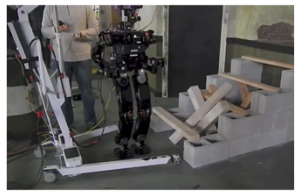
바다에서 원자로까지...











조선해양공학
부유체 동역학
Dynamic Positioning

기계공학
무인 자동차

항공공학
GNC, GPS-denied 환경
SLAM + 비선형 최적화

SHI
생산자동화 로봇
O&M 로봇

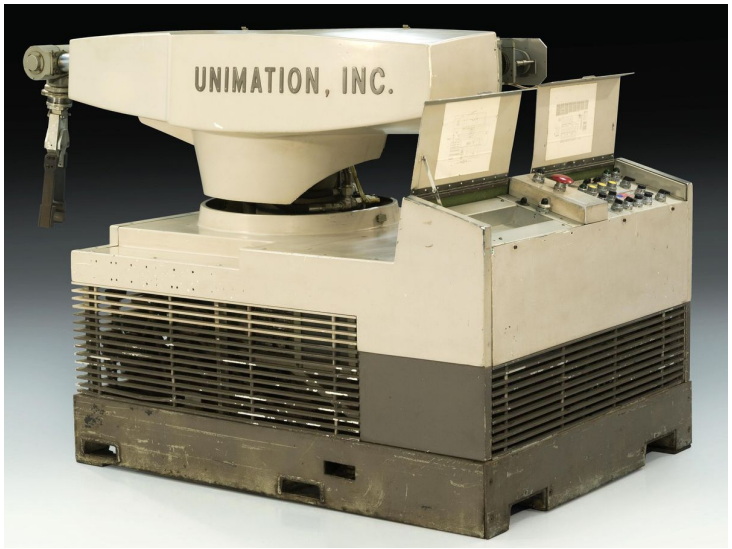
GIT.
DARPA
Robotic Challenge

KAERI
원자력 로봇

로봇 기술의 역사

왜 원자력연구원 연구자가 로봇기술을 연구할까?

문헌



In 1961, the First Robot Arm Punched In The era of industrial robots began with **Unimate**, at a GM plant in Trenton



3,500 pound industrial robot, **Unimate**, pouring coffee for a woman at Biltmore Hotel in Los Angeles, Calif., 1967.



SRI researchers Nils Nilsson (right) and Sven Wahlstrom with **Shakey** the Robot in the late 1960s

원자력에서 발전한 master–slave 원격조작

문헌

왜 원자력연구원 연구자가 로봇기술을 연구할까?



Goertz demonstrating his mechanical slave-master manipulator device in 1949 at Argonne National Lab.

master–slave 원격조작 · ANL (Goertz), 1949
차폐창 너머에서 방사성 물질을 다루기 위해
손의 움직임과 도구의 움직임을 연동

힘 반영 (force reflection)	전기식 서보 (1954, Model E1) 로봇??	감독 제어 (supervisory control)

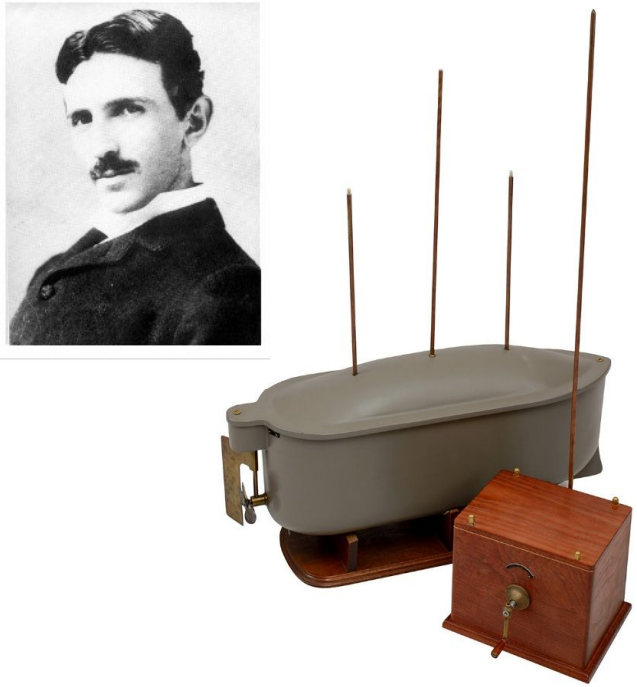
기술이 확산된 사례

우주	수중	의료

잠깐~

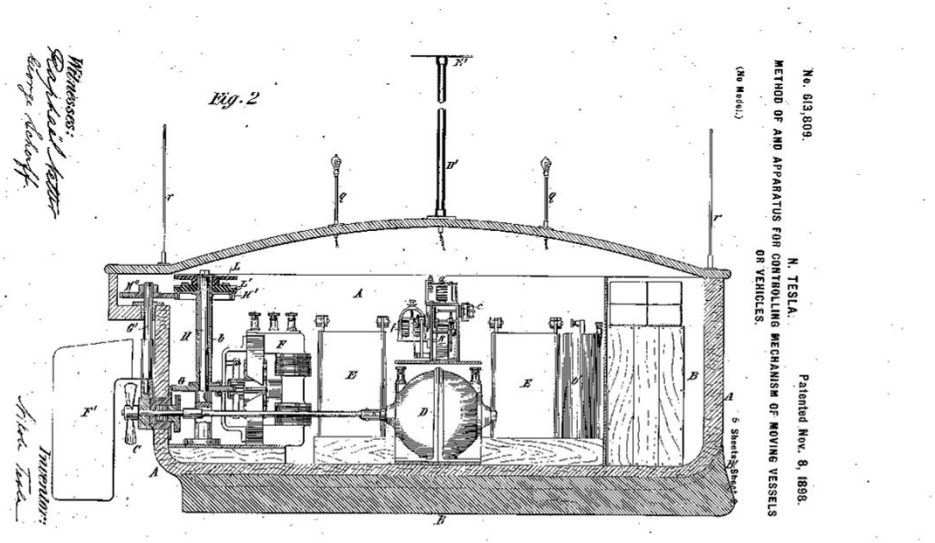
해양로봇의 역사?

문헌



Tesla;s Remotely Controlled Boat (Telautomaton) @ Electrical Exhibition at Madison Square Garden, NY, 1898

telautomatics - the art of controlling the movements and operations of distant automatons



Method of and apparatus for controlling mechanism of moving vessels or vehicles (US Patent 613,809, N. Tesla, 1898)

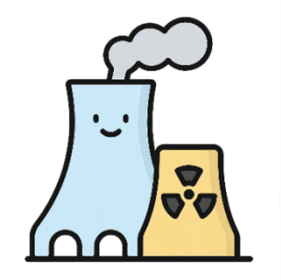
바다에서 → **원자로까지** → 다시 바다로

KAERI 로봇 연구













KAERI 로봇 연구



Pipe inspection



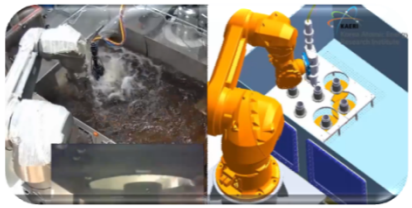
RPV inspection



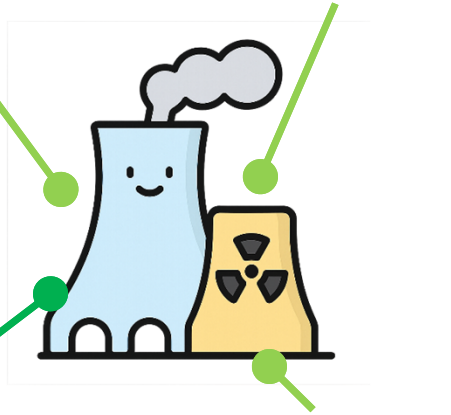
Confined space monitoring



Dismantling manipulator



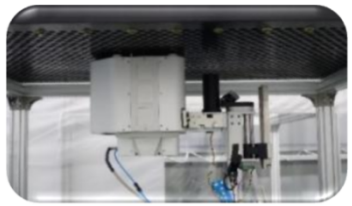
Remote cutting technology

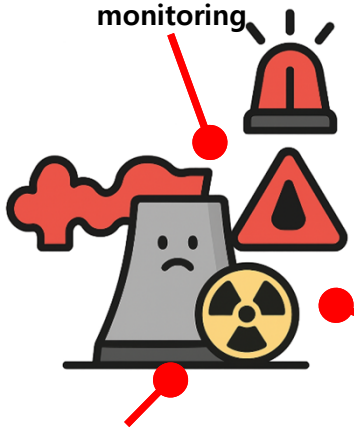


Spent fuel inspection



Steam generator inspection





Indoor monitoring



Outdoor monitoring

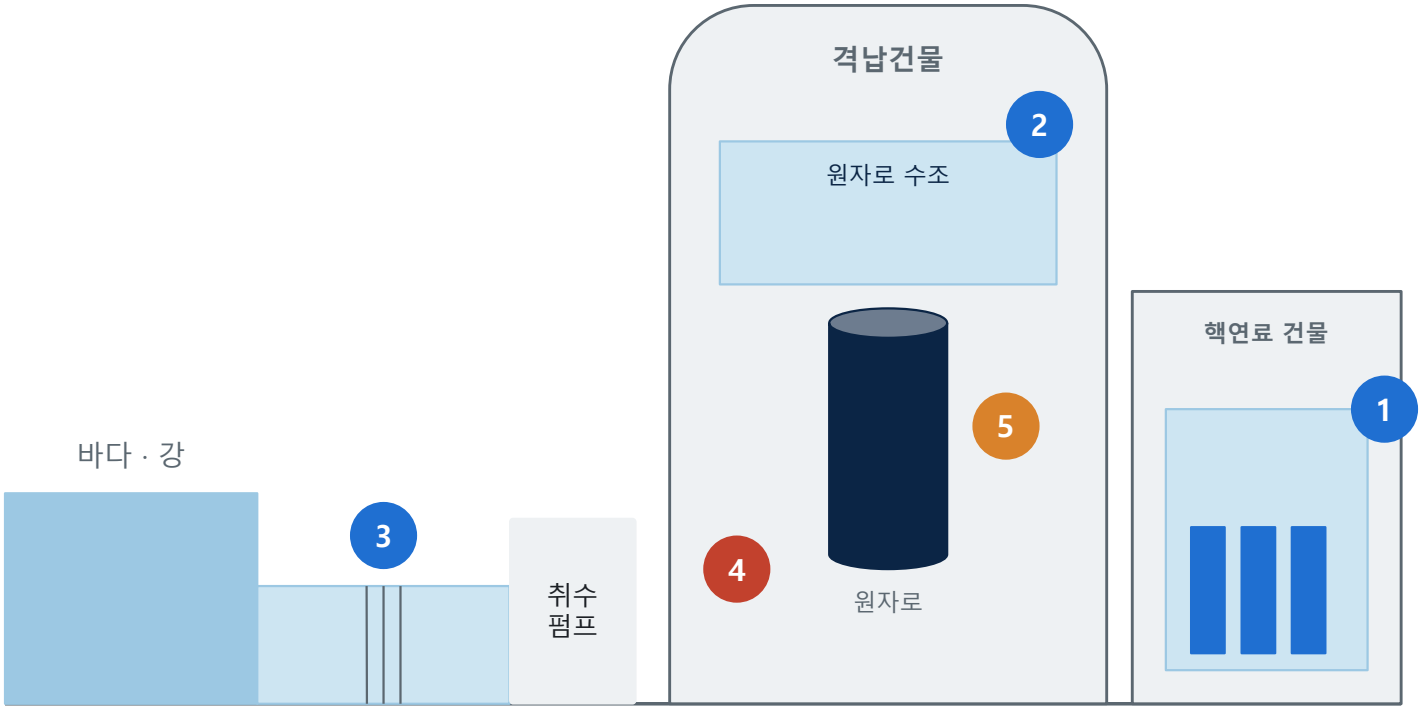




Heavy payload mobile manipulation



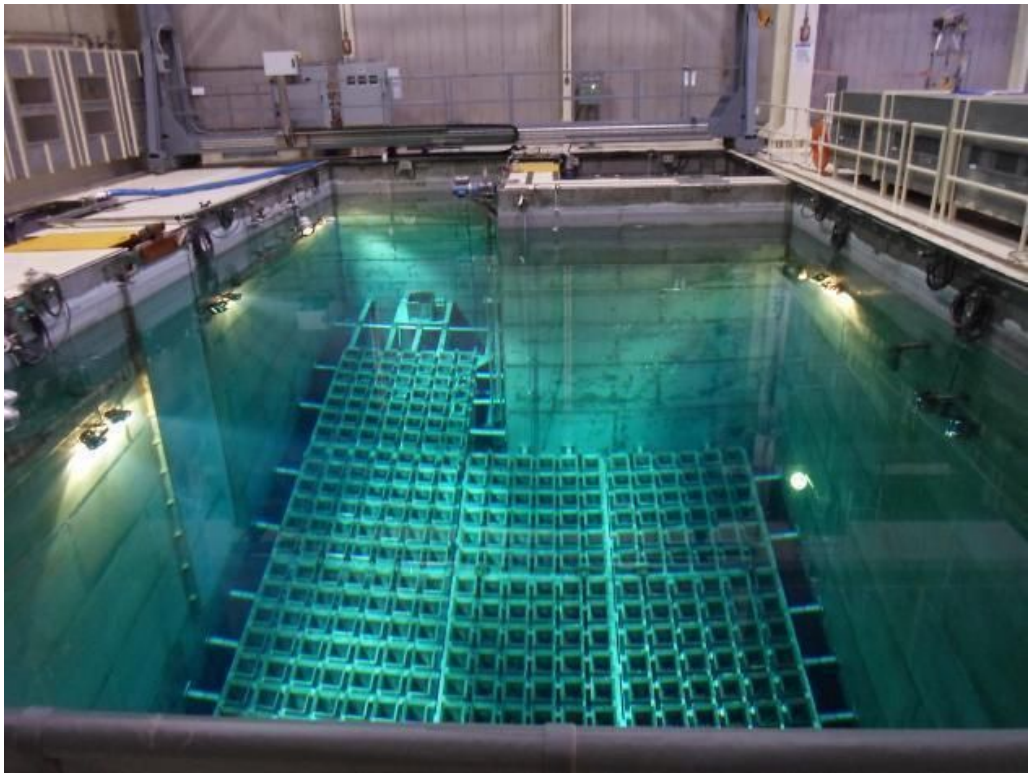
원자력 발전소 주변의 '물'

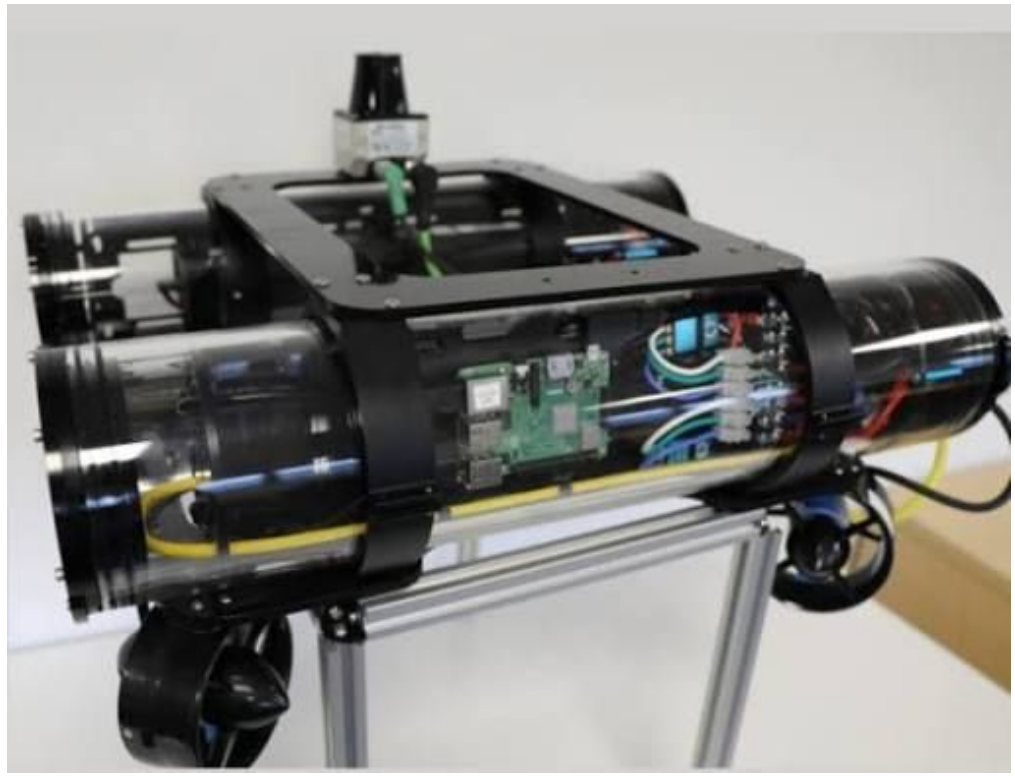


1. **사용후핵연료 저장조**
   IAEA 안전조치 · 점검
2. **원자로 수조 · 원자로 내부**
   정비 기간 수중 검사
3. **취수구 · 방수로**
   취수 구조물 · 트래시랙 점검
4. **사고 원자로 격납용기**
   침수 내부 조사 (후쿠시마)
5. **해체 현장**
   원자로 구조물 수중 절단

사용후 연료 저장조

문헌

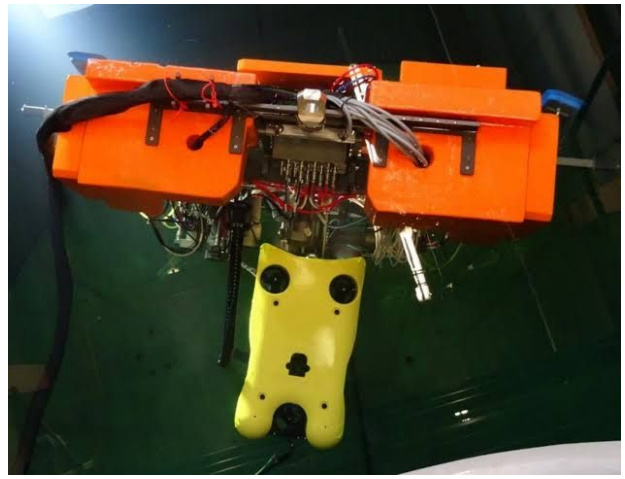


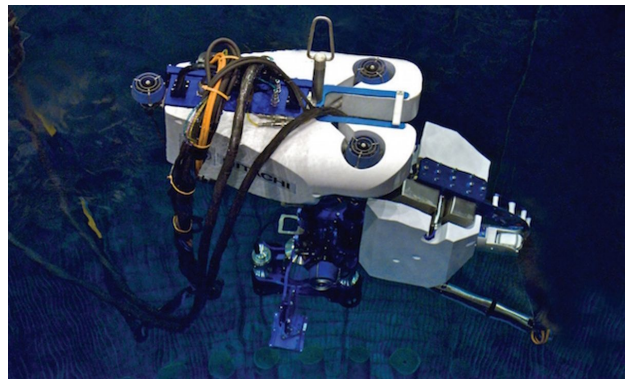


원자로 수조

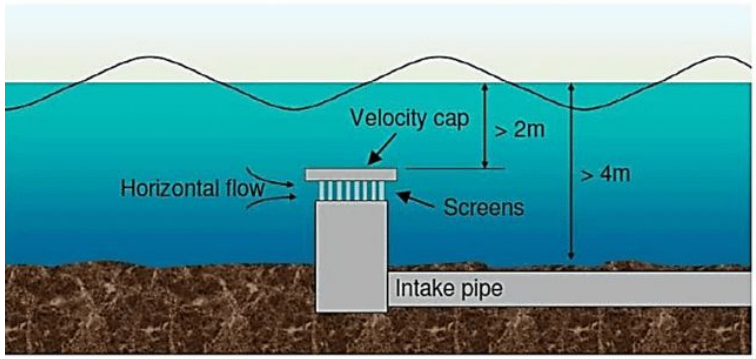


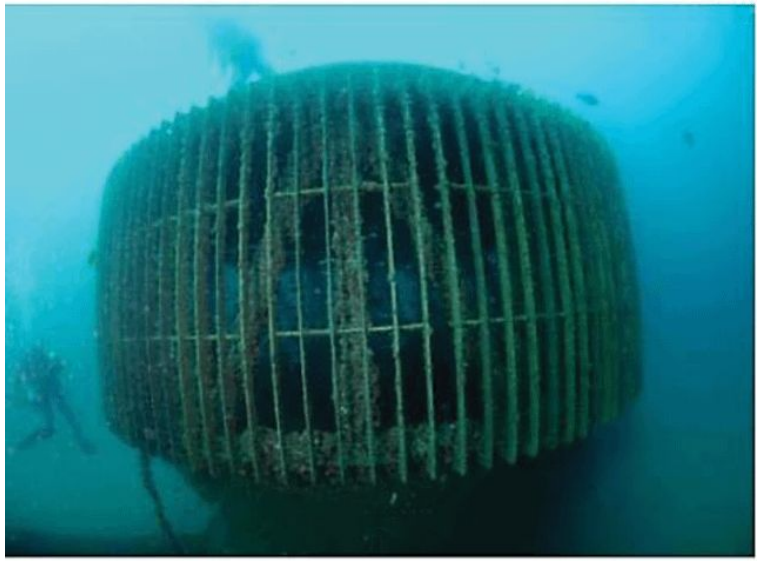






원자력 발전소 취수구



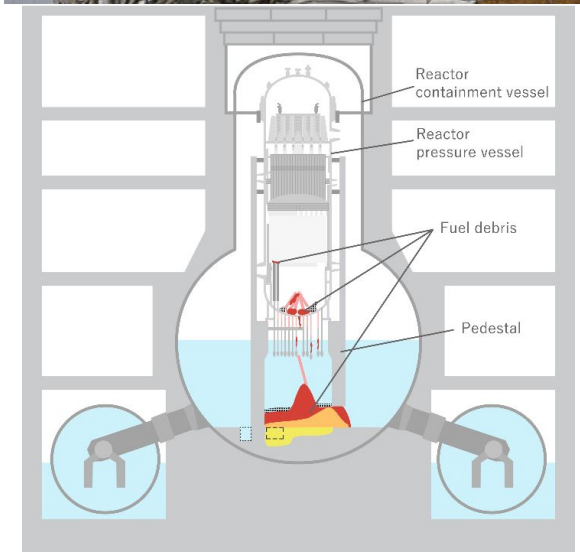




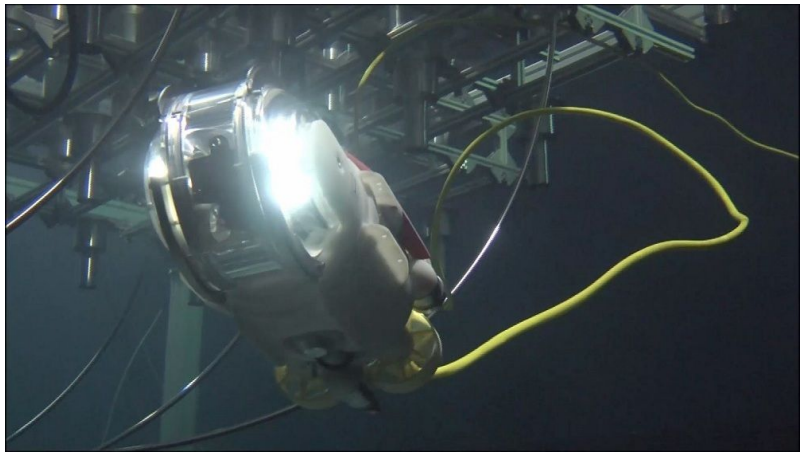


원자력 발전소 사고현장




Reactor containment vessel
Reactor pressure vessel
Fuel debris
Pedestal

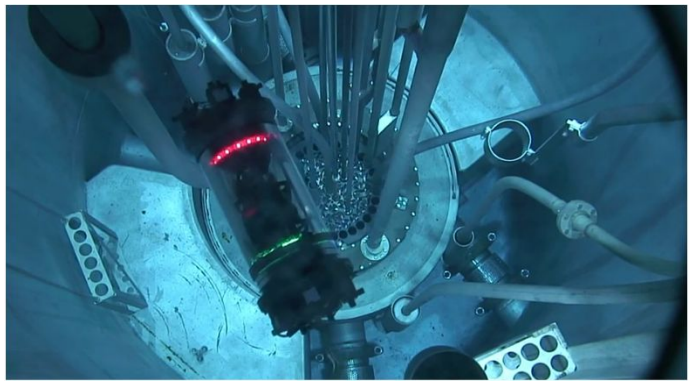




노후 원자력 시설





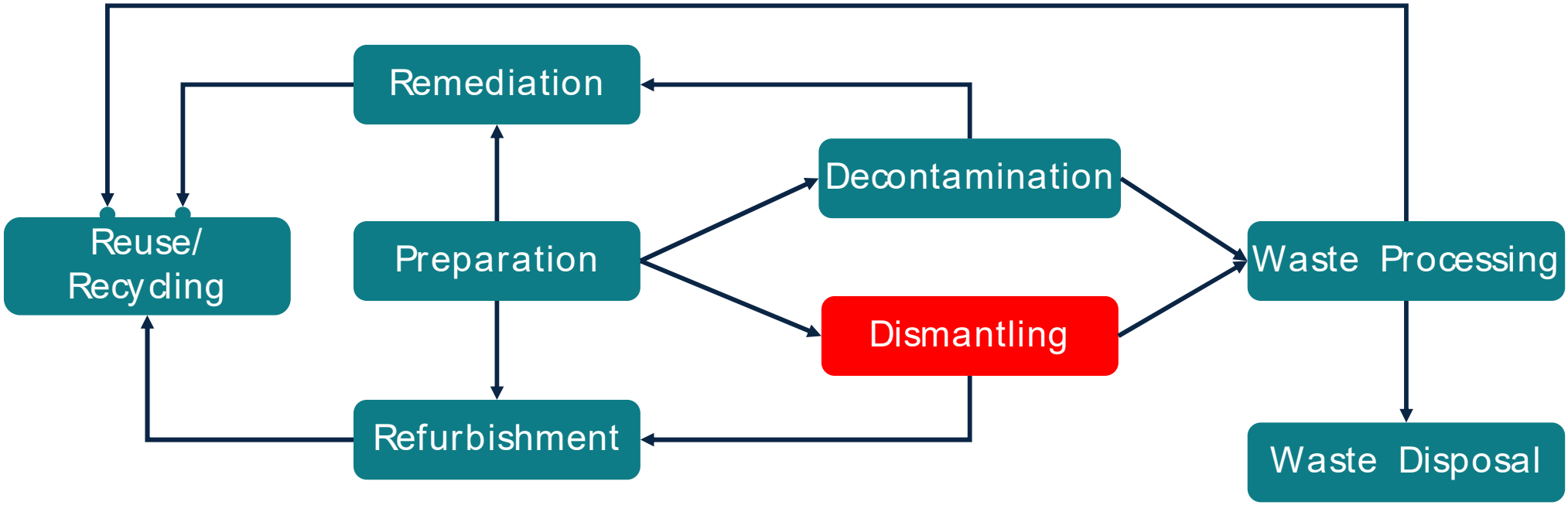


원자력 시설의 해체

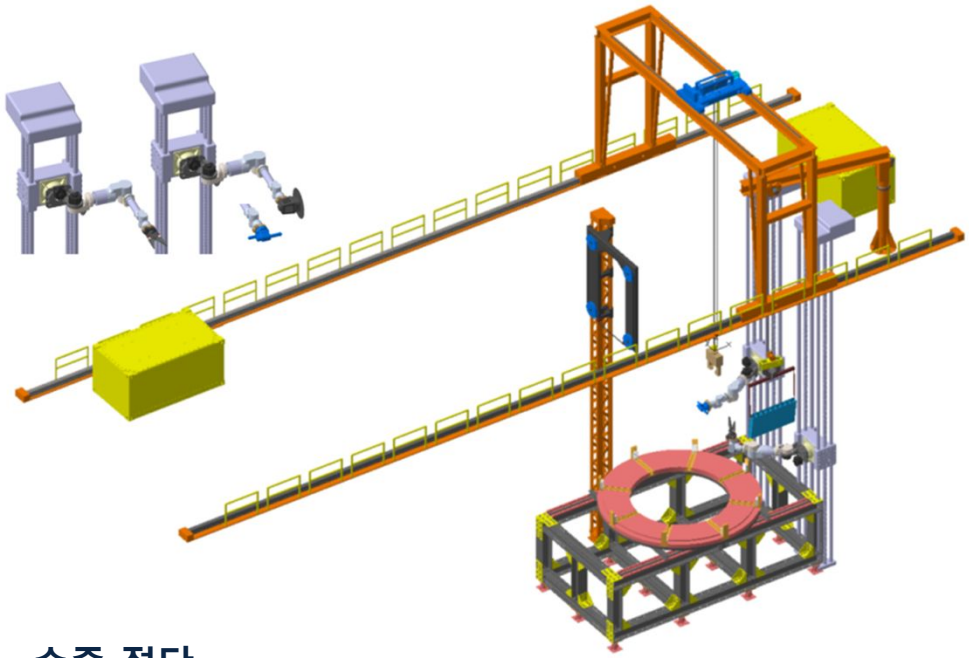


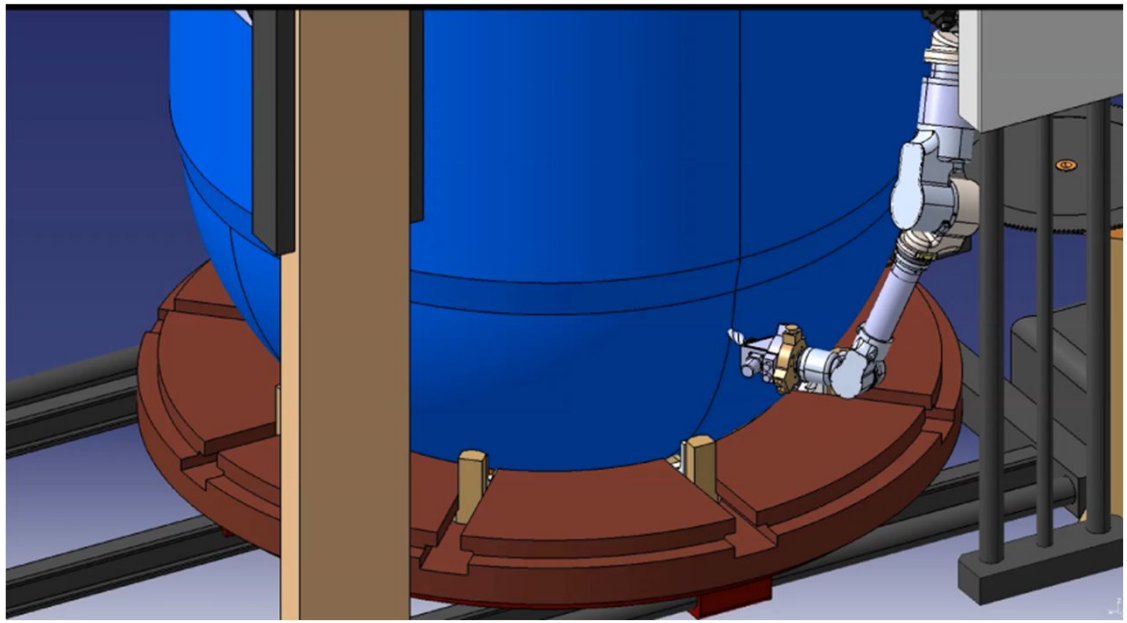


원자력 시설의 해체


Remediation
Decontamination
Reuse/
Recycling
Preparation
Waste Processing
Dismantling
Refurbishment
Waste Disposal

해체로봇 시스템

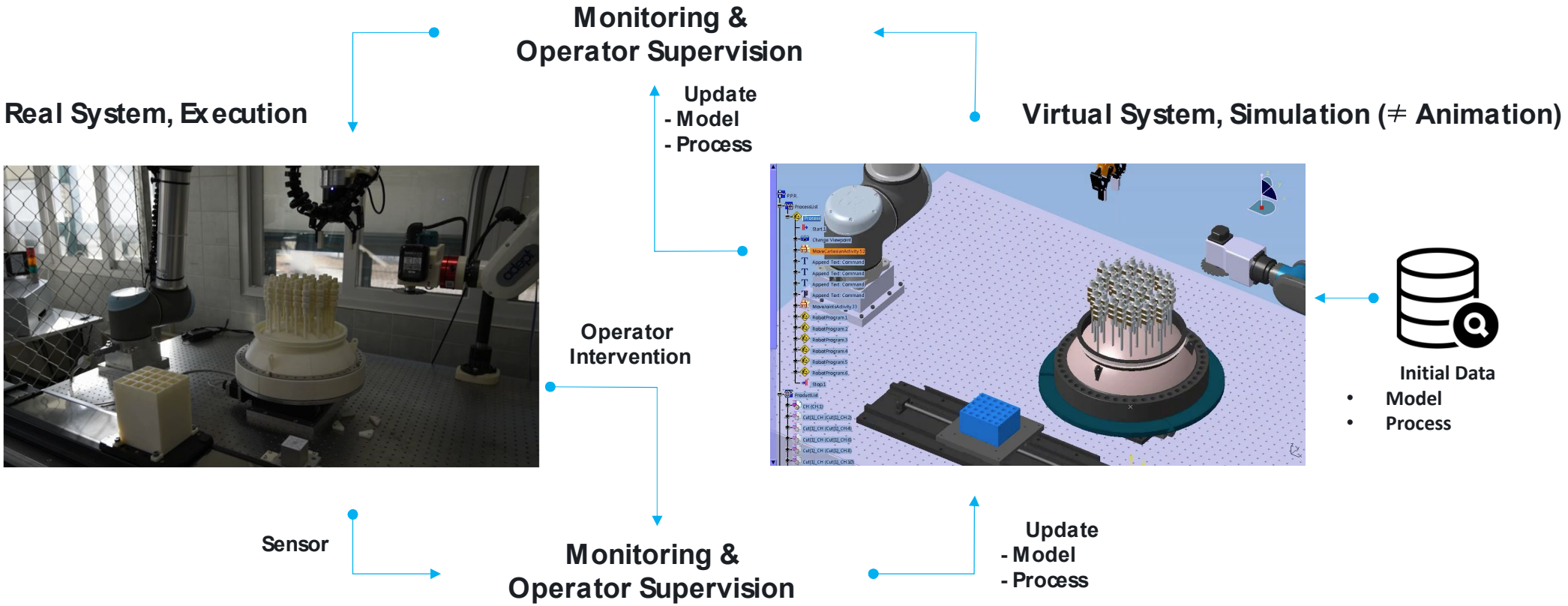




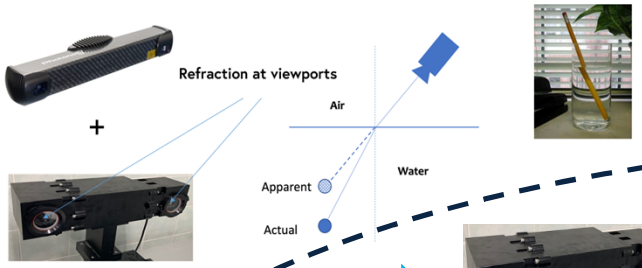
수중 절단

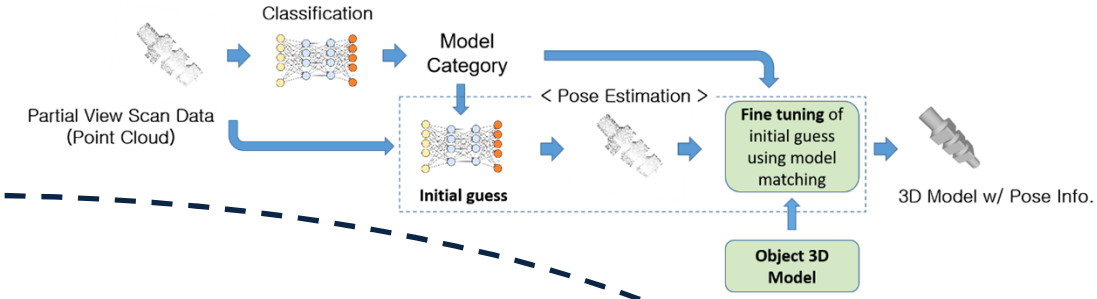
- 대상: 고방사화된 원자력 구조물 (예. 원자로, 1차계통)
- 장점: 물이 방사선을 차폐하고, 공기 중 입자 비산을 줄임
- 단점: 수중 오염 확산과 여과 관리가 함께 필요

해체로봇 시스템


Monitoring &
Operator Supervision
Real System, Execution
Update
- Model
- Process
Virtual System, Simulation (≠ Animation)
Operator
Intervention
Initial Data
• Model
• Process
Sensor
Monitoring &
Operator Supervision
Update
- Model
- Process

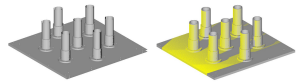
해체로봇 시스템



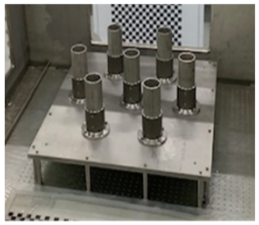




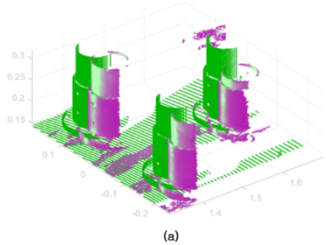
Distortion

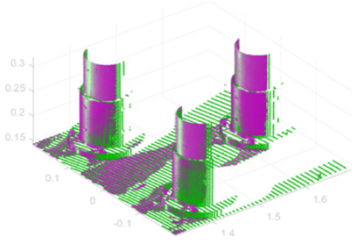


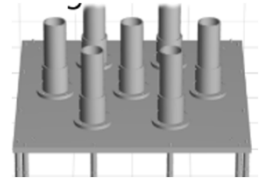
Classification, Pose Est.



Physical System



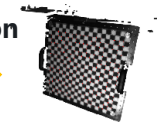


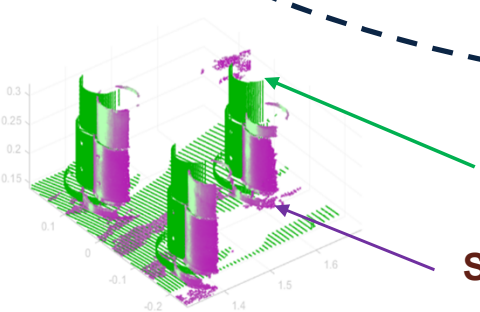


3D Model Reconstruction (Virtual System)



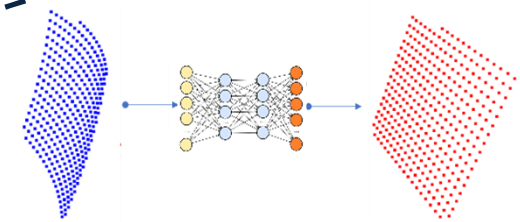
Correction





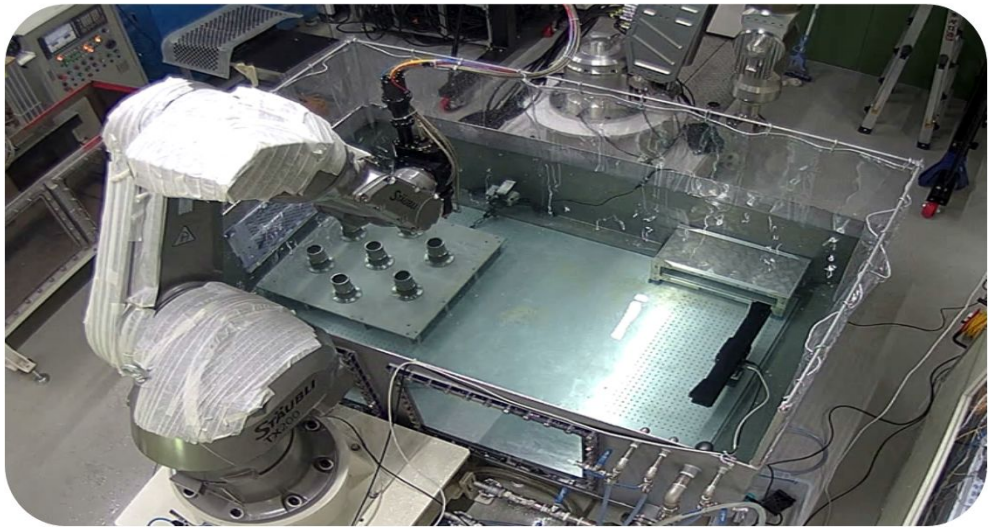
Scanned in the water

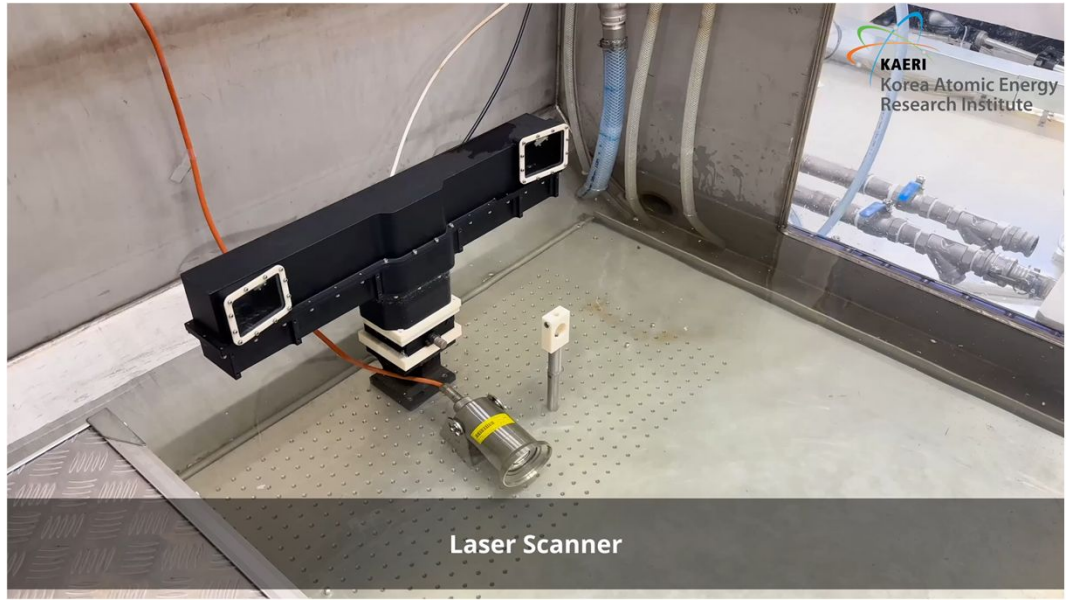
Scanned in the air



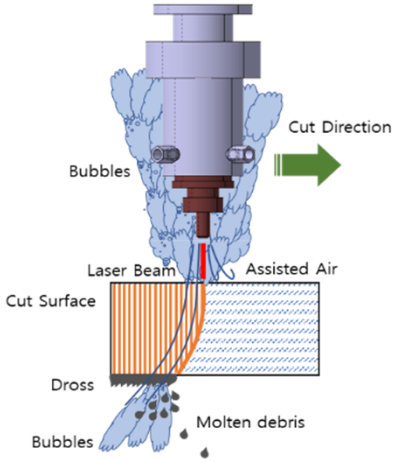
Correction Model

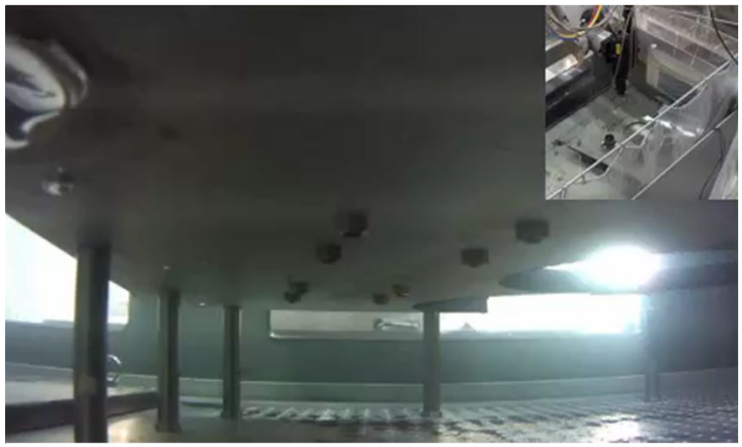
해체로봇 시스템




KAERI
Korea Atomic Energy
Research Institute
Laser Scanner

해체로봇 시스템

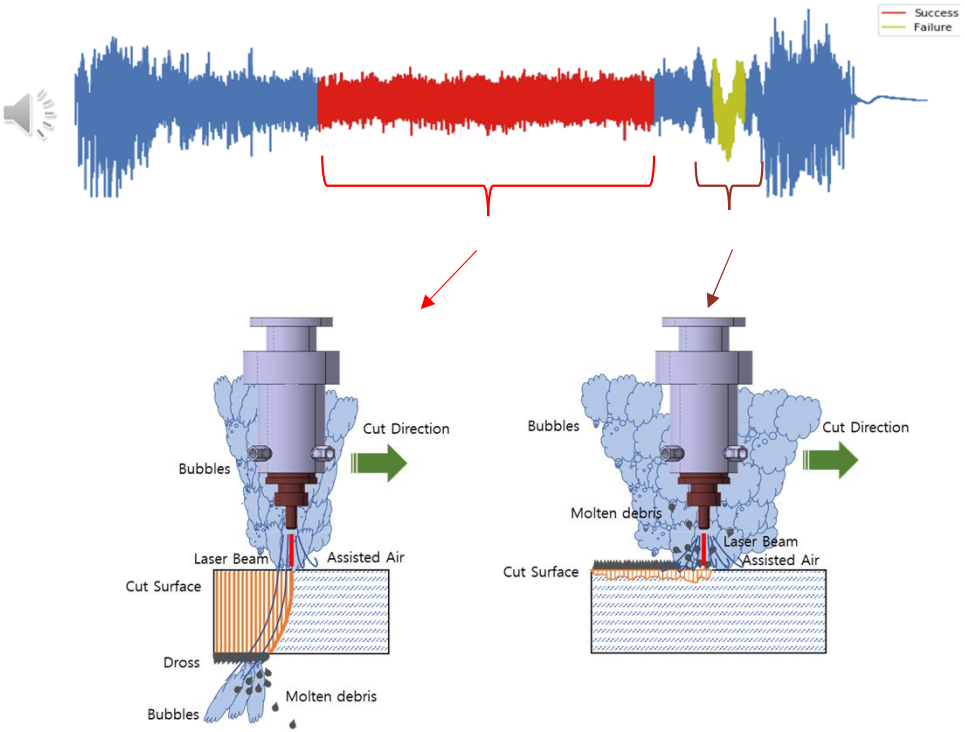


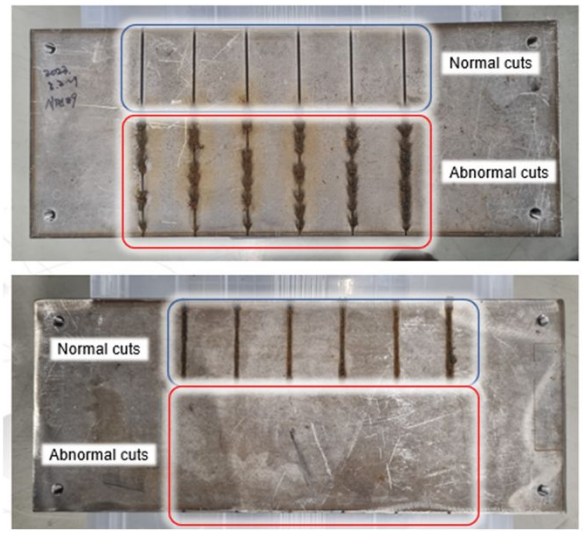




Video **Vibration** **Temperature** **Pressure** **Sound** ...

해체로봇 시스템



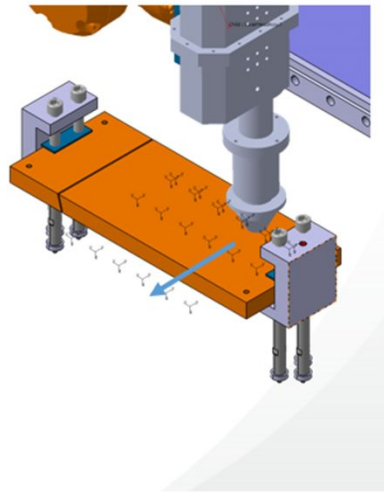


150 mm/min

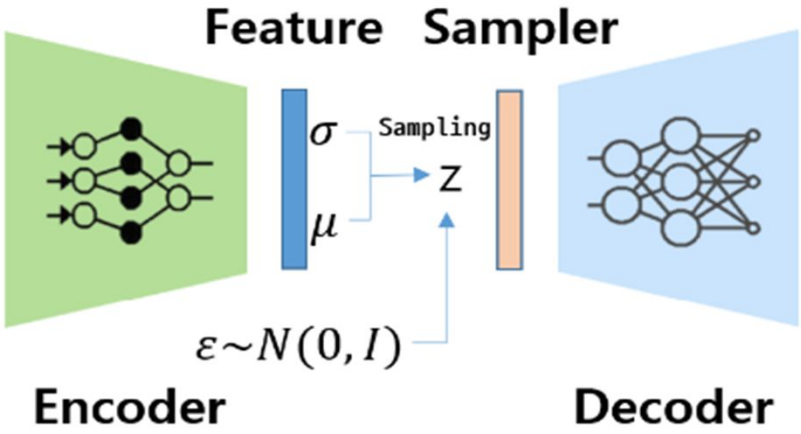
Top

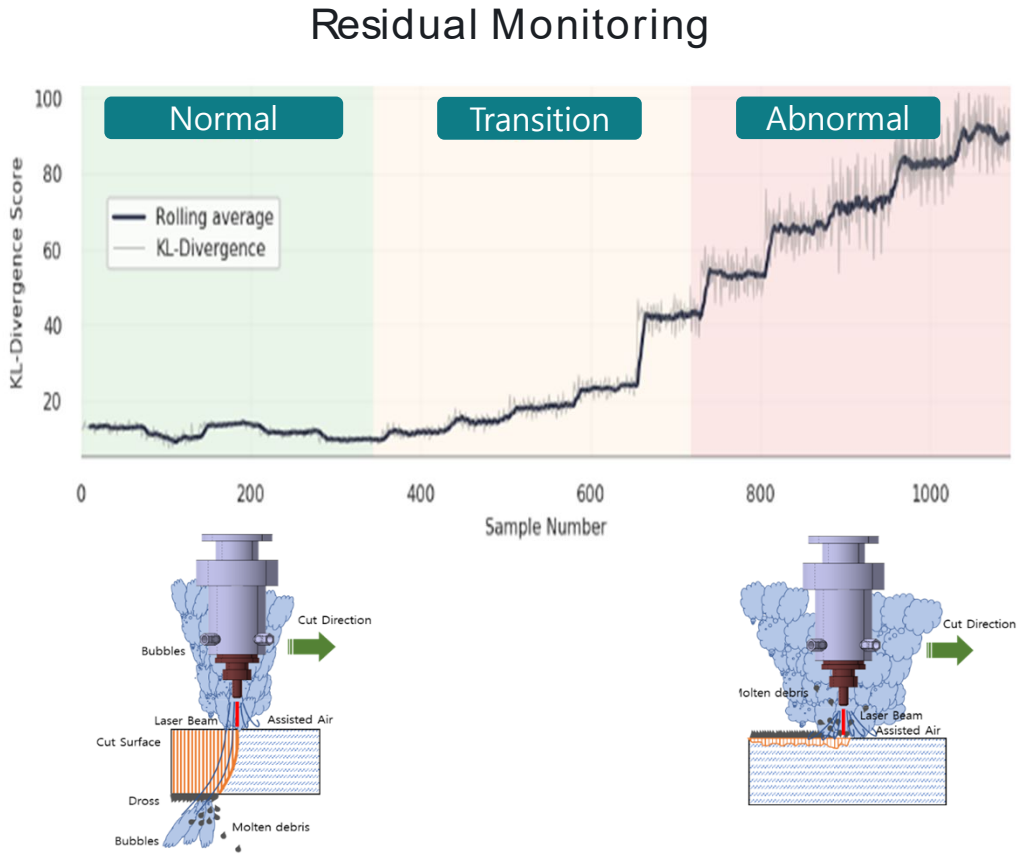
400 mm/min

Bottom

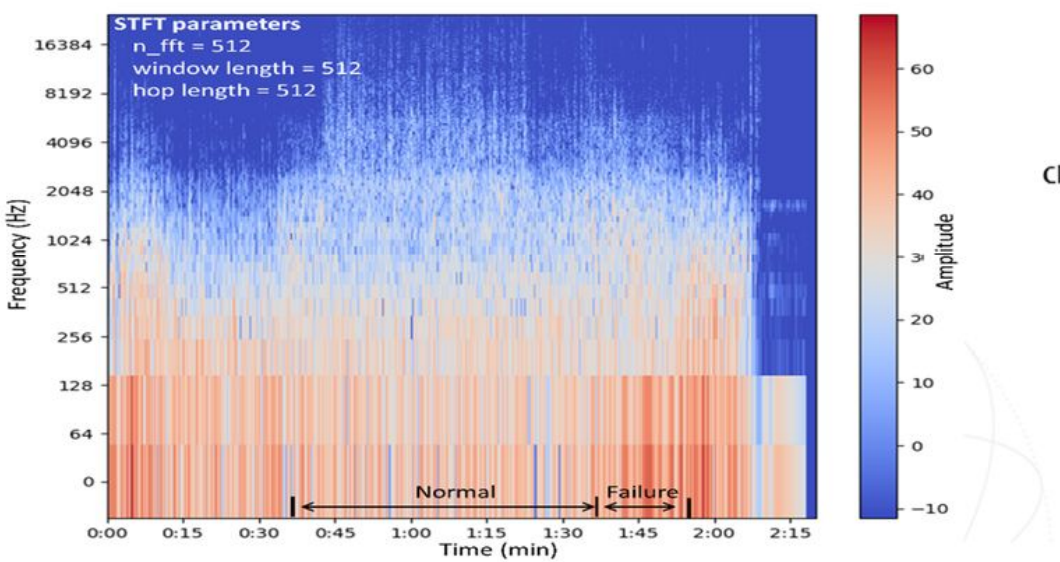


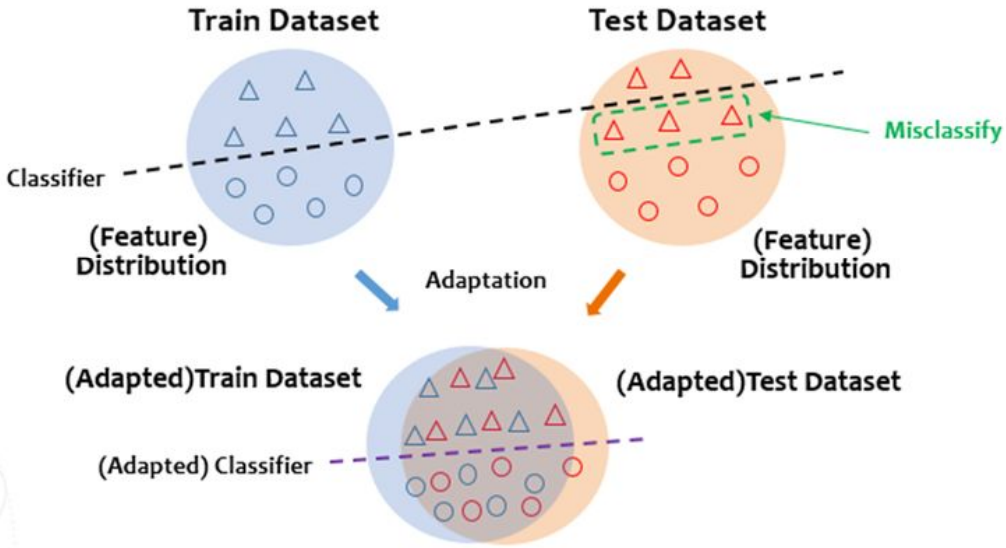
해체로봇 시스템


Feature
Sampler
σ
μ
Sampling
Z
ε~N(0, I)
Encoder
Decoder


Residual Monitoring
Normal
Transition
Abnormal
Rolling average
KL-Divergence
KL-Divergence Score
Sample Number
100
80
60
40
20
0
200
400
600
800
1000
Cut Direction
Bubbles
Laser Beam
Assisted Air
Cut Surface
Dross
Molten debris
Bubbles
Cut Direction
Molten debris
Laser Beam
Assisted Air

해체로봇 시스템





상태 인식의 불확실성 존재함 → 안전성을 보장할 수 있는 설계 필요

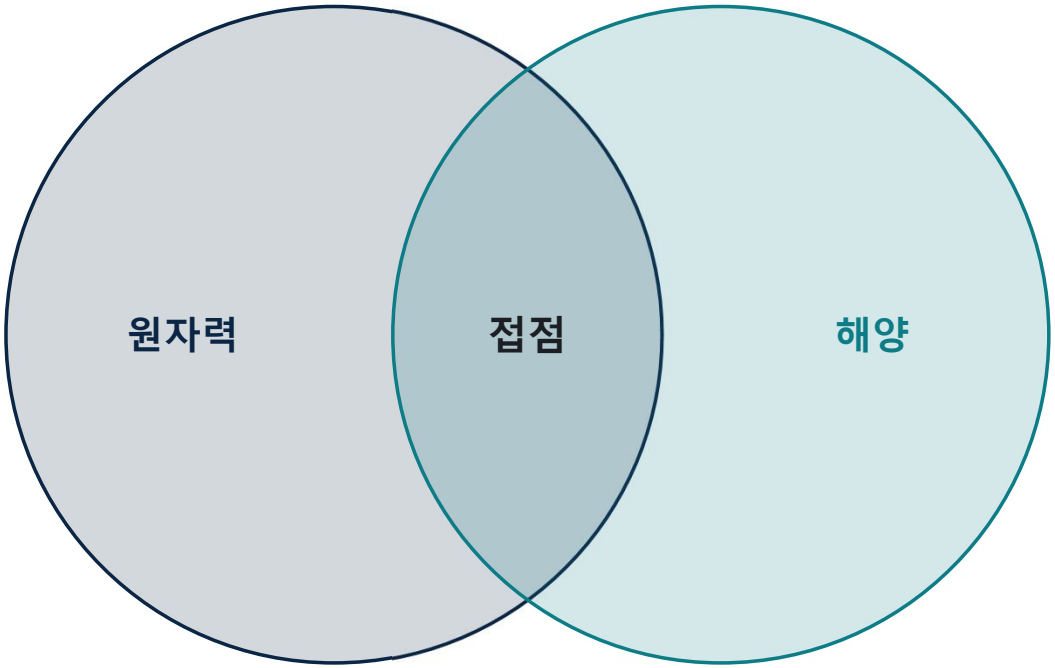
원자력 로봇의 운용 원칙

사람의 단계별 감독/책임하에 원격·자동화 기능을 임무에 맞게 구성

원격·자율화를 제약하는 요인들 …

- **신뢰성**: 고장이 곧 작업 중단·오염 복구/회수 어려움
- **현장환경**: 현장 데이터 수집 어려움 환경인식 어려움
- **작업절차**: 도구 운용· 취급 노하우 필수
- **검증/규제**: 무엇을 어떻게 입증할 것인가
- **경제성**: 개발·시험 비용 대비 효과

서로 다른 현장의 접점



1. 사람의 접근 제약 - 접근
2. (제한된) 자율제어 ·원격(테더)제어 - 운용
3. 시야 · 통신 제약 – 관측/인식
4. 높은 실패/회수 비용 – 복구/경제성
5. 적은 실환경 시험 기회 – 접근/경제성

• • •

마무리

원자력 시설에는 해양로봇 기술을 바로 적용할 수 있는 다양한 수중/수상 작업이 있다.

두 분야는 환경적 측면에서 일부 차이점이 있으나 공통 접점에서 함께 고민해서 해결할 수 있는 문제들이 많다.

